MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – PERIODIC TEST 11
Chapter 13: Statistics

Time: 60 Minutes
MM: 30

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided in question subparts.
2. **Section A** comprises 5 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer (VSA) questions of 2 marks each.
4. **Section C** comprises 2 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 2 Long Answer (LA) questions of 5 marks each.
6. **Section E** comprises 1 Case Study based question of 4 marks with internal subparts.
7. Use of calculators is not permitted. All rough work and calculation steps must be clearly shown.

SECTION A: Multiple Choice & Assertion-Reason (6 × 1 = 6 Marks)

Q1. If $u_i = \dfrac{x_i - 25}{10}$, $\sum f_i u_i = 20$, and $\sum f_i = 100$, then the value of the mean $\bar{x}$ is: [1]

(A) 27 (B) 25 (C) 30 (D) 35

Q2. For the following data: [1]

Marks	0–10	10–20	20–30	30–40
Students	4	9	15	12

The modal class and median class respectively are:

(A) $20-30$ and $20-30$ (B) $10-20$ and $20-30$
(C) $20-30$ and $30-40$ (D) $30-40$ and $20-30$

Q3. If $x_1, x_2, \ldots, x_n$ are n observations with mean $\bar{x}$, then the mean of $ax_1 + b, ax_2 + b, \ldots, ax_n + b$ is: [1]

(A) $a\bar{x}$ (B) $\bar{x} + b$
(C) $a\bar{x} + b$ (D) $\frac{\bar{x}+b}{a}$

Q4. If the mode of a series exceeds its mean by 12, then the mode exceeds the median by: [1]

(A) 4 (B) 8 (C) 6 (D) 10

Q5. The class mark of the class $90 - 120$ is: [1]

(A) 90 (B) 105 (C) 115 (D) 120

Q6. **Assertion (A):** The mode of a grouped distribution can be calculated directly even when class intervals are discontinuous $(1-10, 11-20)$.
Reason (R): The lower limit (l) and class width (h) in the mode formula must be determined from continuous true class limits. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (2 × 2 = 4 Marks)

Q7. Find the median of all prime numbers less than 20. [2]

Q8. The mean of 5 numbers is 18. If one number is excluded, their mean becomes 16. Find the excluded number. [2]

SECTION C: Short Answer Type (2 × 3 = 6 Marks)

Q9. Find the median height of 60 students from the following data: [3]

Height (cm)	135–140	140–145	145–150	150–155	155–160	160–165
Students	4	9	18	16	10	3

Q10. Find the mean of the following distribution using Assumed Mean method ($a = 32.5$): [3]

Class	15–20	20–25	25–30	30–35	35–40	40–45
Frequency	4	6	10	13	5	2

SECTION D: Long Answer Type (2 × 5 = 10 Marks)

Q11. Find the mean and mode of the following distribution of daily pocket allowance of 64 children: [5]

Allowance (Rs)	11–13	13–15	15–17	17–19	19–21	21–23	23–25
Children	7	6	9	13	20	5	4

Q12. The following distribution gives the state-wise teacher-student ratio in higher secondary schools of India. Find the mode and mean of this data: [5]

Ratio	15–20	20–25	25–30	30–35	35–40	40–45	45–50	50–55
States	3	8	9	10	3	0	0	2

SECTION E: Case Study Based Question (1 × 4 = 4 Marks)

Q13. CASE STUDY: RURAL LIVELIHOOD MISSION SELF-HELP GROUP WEEKLY SAVINGS [4 Marks]

A microfinance cooperative audited the weekly deposit savings (in Rupees) of 120 women members of rural self-help groups (SHGs) under a financial inclusion scheme.

Weekly Savings (Rs)	50–100	100–150	150–200	200–250	250–300
SHG Members	15	25	40	30	10

(i) Calculate the median weekly savings of the cooperative members. [2]

(ii) Identify the modal savings bracket. [1]

(iii) Find the cumulative frequency of members saving less than Rs 200 weekly. [1]

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★